

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Week No.: 3

Student Name: B.Harsha Vardhan

Roll No.: A24126552075

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1. TITLE

Write a Program Using JavaScript (Arrays, Functions)

2. AIM

To write JavaScript programs demonstrating array manipulation methods, custom functions, and ES6 class objects.

3. OBJECTIVE

The objectives of this experiment are:

- Practice JS array functions (filter, reduce).
- Implement ES6 class syntax.

4. THEORY

JavaScript provides array helper functions (filter, reduce, map) and ES6 class syntax for OOP.

5. ALGORITHM / PROCEDURE

1. Create function.js and object.js.
2. Implement score processing with filter and reduce.
3. Define Book class.
4. Run using Node.js.

6. PROGRAM

function.js

```
// Student Score Processing in JavaScript
```

```
let scores = [78, 85, 62, 90, 94];
```

```
function printScores(arr) {  
    console.log("Scores:", arr);  
}
```

```
function calculateSum(arr) {  
    let sum = 0;  
    for (let i = 0; i < arr.length; i++) {  
        sum += arr[i];  
    }  
    return sum;  
}
```

```
function calculateAverage(arr) {
    return calculateSum(arr) / arr.length;
}

function addScore(arr, newScore) {
    arr.push(newScore);
}

printScores(scores);
console.log("Total Score:", calculateSum(scores));
console.log("Average Score:", calculateAverage(scores).toFixed(2));

console.log("\nAdding new score: 88");
addScore(scores, 88);
printScores(scores);
console.log("New Average Score:", calculateAverage(scores).toFixed(2));
```

object.js

```
// Student Course System in JavaScript (ES6 Class)
```

```
console.log("==== Student Management System =====");
```

```
class CourseManager {
    constructor() {
        this.courses = [];
    }

    addCourse(title, credits) {
        this.courses.push({ name: title, points: credits });
        console.log(`Course "${title}" enrolled successfully.`);
    }

    listCourses() {
        console.log("\nEnrolled Courses:");
        for (let i = 0; i < this.courses.length; i++) {
            console.log(`${i + 1}. ${this.courses[i].name} (${this.courses[i].points} Credits)`);
        }
    }

    totalCredits() {
        let total = 0;
        for (let i = 0; i < this.courses.length; i++) {
            total += this.courses[i].points;
        }
        return total;
    }

    removeLastCourse() {
        let dropped = this.courses.pop();
        console.log(`\nDropped Course: ${dropped.name}`);
    }
}

const manager = new CourseManager();
manager.addCourse("Full Stack Web Dev", 4);
manager.addCourse("Machine Learning", 3);
manager.addCourse("Database Systems", 3);
```

```
manager.listCourses();
console.log("\nTotal Credits:", manager.totalCredits());

manager.removeLastCourse();
console.log("\nUpdated List:");
manager.listCourses();
console.log("Updated Total Credits:", manager.totalCredits());
```

7. CODE EXPLANATION

Code / Syntax	Explanation
scores.filter(...)	Filters array elements
scores.reduce(...)	Aggregates array elements
class Book { ... }	ES6 OOP class

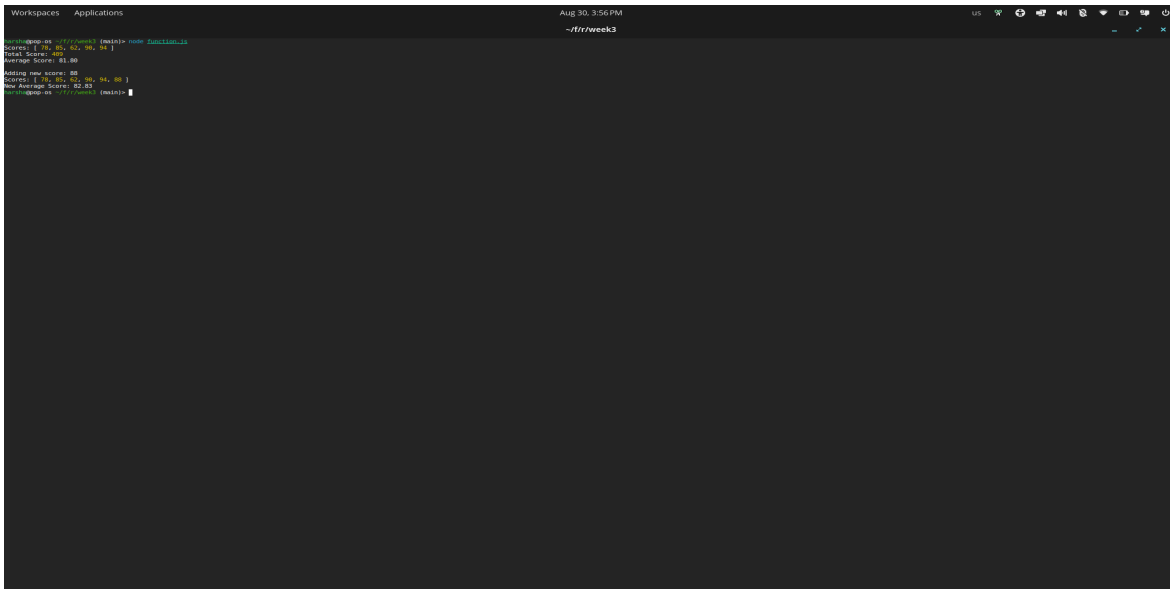
8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Node.js Execution	Console prints processed outputs	Console prints processed outputs	Pass

9. OUTPUT

Output 1: Execution Screenshot



```
Workspaces Applications Aug 30, 3:56 PM
~H1x7week3

node > manager.listCourses()
Courses: [ { id: 101, title: 'JS', avg: 85 }, { id: 102, title: 'Java', avg: 90 }, { id: 103, title: 'Python', avg: 88 } ]
Total Credits: 261
Average Score: 87.67

node > manager.removeLastCourse()
Updated List:
Courses: [ { id: 101, title: 'JS', avg: 85 }, { id: 102, title: 'Java', avg: 90 } ]
Total Credits: 251
Average Score: 87.5
```

10. RESULT

Thus, the experiment for Week 3 (Write a Program Using JavaScript (Arrays, Functions)) was successfully executed and verified.